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Rotation Matrix	Translation Matrix	Offset Matrix	Test Vector
$R := \begin{pmatrix} \textcolor{red}{r}_{11} & r_{12} & 0 \\ r_{21} & r_{22} & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$T := \begin{pmatrix} 1 & 0 & \textcolor{red}{t}_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{pmatrix}$	$T_{\text{ofs}} := \begin{pmatrix} 1 & 0 & -\textcolor{red}{t}_x \\ 0 & 1 & -t_y \\ 0 & 0 & 1 \end{pmatrix}$	$V := \begin{pmatrix} \textcolor{red}{x} \\ y \\ 1 \end{pmatrix}$

Rotate a vector 2x2

$$R \cdot V \rightarrow \begin{pmatrix} r_{11} \cdot x + r_{12} \cdot y \\ r_{21} \cdot x + r_{22} \cdot y \\ 1 \end{pmatrix}$$

Translate a vector 2x2

$$T \cdot V \rightarrow \begin{pmatrix} \textcolor{red}{t}_x + x \\ t_y + y \\ 1 \end{pmatrix}$$

Rotate than translate

$$T \cdot R \rightarrow \begin{pmatrix} r_{11} & r_{12} & \textcolor{red}{t}_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{pmatrix}$$

Translate than rotate

$$R \cdot T \rightarrow \begin{pmatrix} r_{11} & r_{12} & r_{11} \cdot \textcolor{red}{t}_x + r_{12} \cdot t_y \\ r_{21} & r_{22} & r_{21} \cdot \textcolor{red}{t}_x + r_{22} \cdot t_y \\ 0 & 0 & 1 \end{pmatrix}$$

Translate than rotate than translate (rotate around the origin ($\textcolor{red}{t}_x$, t_y))

$$T \cdot R \cdot T_{\text{ofs}} \rightarrow \begin{pmatrix} r_{11} & r_{12} & \textcolor{red}{t}_x - r_{11} \cdot \textcolor{red}{t}_x - r_{12} \cdot t_y \\ r_{21} & r_{22} & t_y - r_{21} \cdot \textcolor{red}{t}_x - r_{22} \cdot t_y \\ 0 & 0 & 1 \end{pmatrix}$$

Rotation Matrix	Translation Matrix	Offset Matrix	Test Vector
$R := \begin{pmatrix} \textcolor{red}{r}_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$	$T := \begin{pmatrix} 1 & 0 & 0 & \textcolor{red}{t}_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}$	$T_{\text{ofs}} := \begin{pmatrix} 1 & 0 & 0 & -\textcolor{red}{t}_x \\ 0 & 1 & 0 & -t_y \\ 0 & 0 & 1 & -t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}$	$V := \begin{pmatrix} \textcolor{red}{x} \\ y \\ z \\ 1 \end{pmatrix}$

Rotate a vector 2x2

$$R \cdot V \rightarrow \begin{pmatrix} r_{11} \cdot x + r_{12} \cdot y + r_{13} \cdot z \\ r_{21} \cdot x + r_{22} \cdot y + r_{23} \cdot z \\ r_{31} \cdot x + r_{32} \cdot y + r_{33} \cdot z \\ 1 \end{pmatrix}$$

Translate a vector 2x2

$$T \cdot V \rightarrow \begin{pmatrix} t_x + x \\ t_y + y \\ t_z + z \\ 1 \end{pmatrix}$$

Rotate than translate

$$T \cdot R \rightarrow \begin{pmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Translate than rotate

$$R \cdot T \rightarrow \begin{pmatrix} r_{11} & r_{12} & r_{13} & r_{11} \cdot t_x + r_{12} \cdot t_y + r_{13} \cdot t_z \\ r_{21} & r_{22} & r_{23} & r_{21} \cdot t_x + r_{22} \cdot t_y + r_{23} \cdot t_z \\ r_{31} & r_{32} & r_{33} & r_{31} \cdot t_x + r_{32} \cdot t_y + r_{33} \cdot t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Translate than rotate than translate (rotate around the origin (t_x, t_y))

$$T \cdot R \cdot T_{\text{ofs}} \rightarrow \begin{pmatrix} r_{11} & r_{12} & r_{13} & t_x - r_{11} \cdot t_x - r_{12} \cdot t_y - r_{13} \cdot t_z \\ r_{21} & r_{22} & r_{23} & t_y - r_{21} \cdot t_x - r_{22} \cdot t_y - r_{23} \cdot t_z \\ r_{31} & r_{32} & r_{33} & t_z - r_{31} \cdot t_x - r_{32} \cdot t_y - r_{33} \cdot t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}$$